

DEVOPS STUDY SYLLABUS

Linux · Networking · Azure · Git/GitHub · Terraform · CI/CD · Monitoring · Security

1. Linux	2. Networking
3. Azure	4. Git / GitHub
5. Terraform	6. CI/CD (GitHub Actions / ADO)
7. Monitoring	8. Security

1 LINUX — FOUNDATION

- File & directory commands
- ls, cd, cp, mv, rm
- grep, find, cat, less
- Permissions: chmod, chown
- Users & groups
- Processes: ps, top, kill
- Services: systemctl
- Package management
- SSH
- Environment variables
- Logs
- Cron jobs
- Bash scripting basics

GOAL SSH karke Linux server manage kar pao.

2 NETWORKING

- OSI basics
- TCP/IP
- IPv4
- Subnetting
- CIDR
- Public vs Private IP
- DNS
- DHCP
- Ports
- TCP vs UDP
- HTTP/HTTPS
- SSL/TLS
- Firewall
- NAT
- Routing
- Load Balancer
- Proxy / Reverse Proxy

GOAL Azure networking se connect karke seekhna.

3 AZURE

3.1 Azure Hierarchy

- Management Group (MG)
- Subscription
- Resource Group
- Resource Association
- Governance
- Security

3.2 IAM

- Users
- Groups
- RBAC

3.3 Governance

- Azure Policy
- Policy Initiatives
- Built-in Policies
- Custom Policies

3.4 Landing Zone

- Resource Groups
- Networking
 - VNet
 - Subnet
 - NSG
 - ASG
 - Firewall
 - VNet Peering
 - Global VNet Peering
 - ExpressRoute
 - VPN Gateway
 - Site-to-Site VPN
 - Point-to-Site VPN

3.5 Compute

- Virtual Machine
- VMSS
- Web Apps
- Azure Functions
- Azure Container Services
- Azure Kubernetes Services (AKS)

3.6 Backup & DR

- VM Backup
- Database Backup
- Storage Backup
- RTO / RPO
- Backup Vaults
- ASR (Azure Site Recovery)

3.7 Database

- Azure SQL Server
- Cosmos DB

3.8 Advanced Networking

- Azure Load Balancer
- Application Gateway
- Azure Front Door
- Traffic Manager

3.9 Security

- WAF
- NSG
- ASG
- Key Vault
- Private Endpoint
- Service Endpoint
- Bastion Host
- Defender
- Sentinel
- DDoS Protection

GOAL Ek complete monolithic application ka Azure infrastructure samajhna.

4 GIT + GITHUB

4.1 Git

- clone
- init
- add
- commit

- push
- pull
- fetch
- merge
- rebase
- stash
- reset
- revert
- Branching
- Tags

4.2 GitHub

- Repository
- Branches
- Pull Request
- Code Review
- Branch Protection / Rulesets
- Secrets
- Environments
- Releases
- GitHub Actions

5 TERRAFORM

Must Master

- Terraform basics
- Terraform commands
- Provider
- Resources
- Variables
- Outputs
- Locals
- Data sources
- AzureRM
- Azure resources
- State
- Remote Backend
- Modules
- count / for_each
- depends_on
- lifecycle
- Import
- Drift
- Terraform + Git
- Terraform + CI/CD
- Checkov / tfsec / TFLint

Basic Understanding Enough

- Complex Terraform functions
- Dynamic blocks
- Workspaces
- Advanced module design

6 CI/CD

6.1 GitHub Actions

- Workflow
- Trigger
- Jobs
- Steps
- Runner
- Actions
- Secrets
- Variables
- Environments
- Artifacts
- Approval
- Deployment

6.2 Azure DevOps

- Organization

- Project
- Repository
- Pipeline
- YAML
- Agent
- Agent Pool
- Variable Groups
- Library
- Service Connection
- Environments
- Approvals
- Artifacts
- Deployment

GOAL *Azure DevOps YAML pipeline ko especially strong kar.*

7 MONITORING

Azure Monitor

- Metrics
- Alerts
- Logs

Log Analytics

- Workspace
- Log queries
- KQL basics

Application Monitoring

- CPU
- Memory
- Disk
- Network
- Application logs
- HTTP response codes
- Availability
- Error rate

Application → Azure Monitor → Log Analytics → Alert → Notification

GOAL *Interview mein explain kar pao: "Application down ho gayi toh main kaise identify aur troubleshoot karunga?"*

8 SECURITY

Azure Security

- RBAC
- Managed Identity
- Key Vault
- NSG
- Azure Policy
- Defender for Cloud basics

Terraform Security

- Checkov
- tfsec
- tflint
- Secret handling
- State security

CI/CD Security

- Secrets
- Service Connections
- Least privilege
- Secret scanning
- Dependency scanning
- SAST basics